



Mixture-of-Prefetchers

A compact L2-cache router tuned by OpenEvolve

- Trace behavior shifts

Best prefetcher changes by workload.
- Route cheaply

Observe one 500K-instruction epoch, choose the next.

- Tune policy

OpenEvolve edits compact routing constants.
- Use one baseline

Prefetcher off is 1.000x in every plot.

What We Built On Athena

- Contribution over Athena

Athena provides the simulator and L2C prefetchers. We add routing, epoch counters, fixed splits, OpenEvolve search, and rebuildable plots.
- Experts: MLOP + SPP+PPF at L2C.
- MoP-V1: one-epoch probe router.
- MoP-V2: OpenEvolve-tuned router.

Evaluation Protocol

- Prefetcher off is the 1.000x baseline.
- Best expert = max(MLOP, SPP+PPF).
- Split: 17 training, 7 heldout.
- Epoch: 500K retired instructions.
- IPC tracks cycles under fixed instructions.

Two Athena Experts

- MLOP

Learns useful address offsets across multiple lookahead distances. Better on 4 of 13 validation traces.
- SPP+PPF

Tracks page signatures and delta paths, then filters candidates with a perceptron. Better on 9 of 13.

Trace Sets

- Official

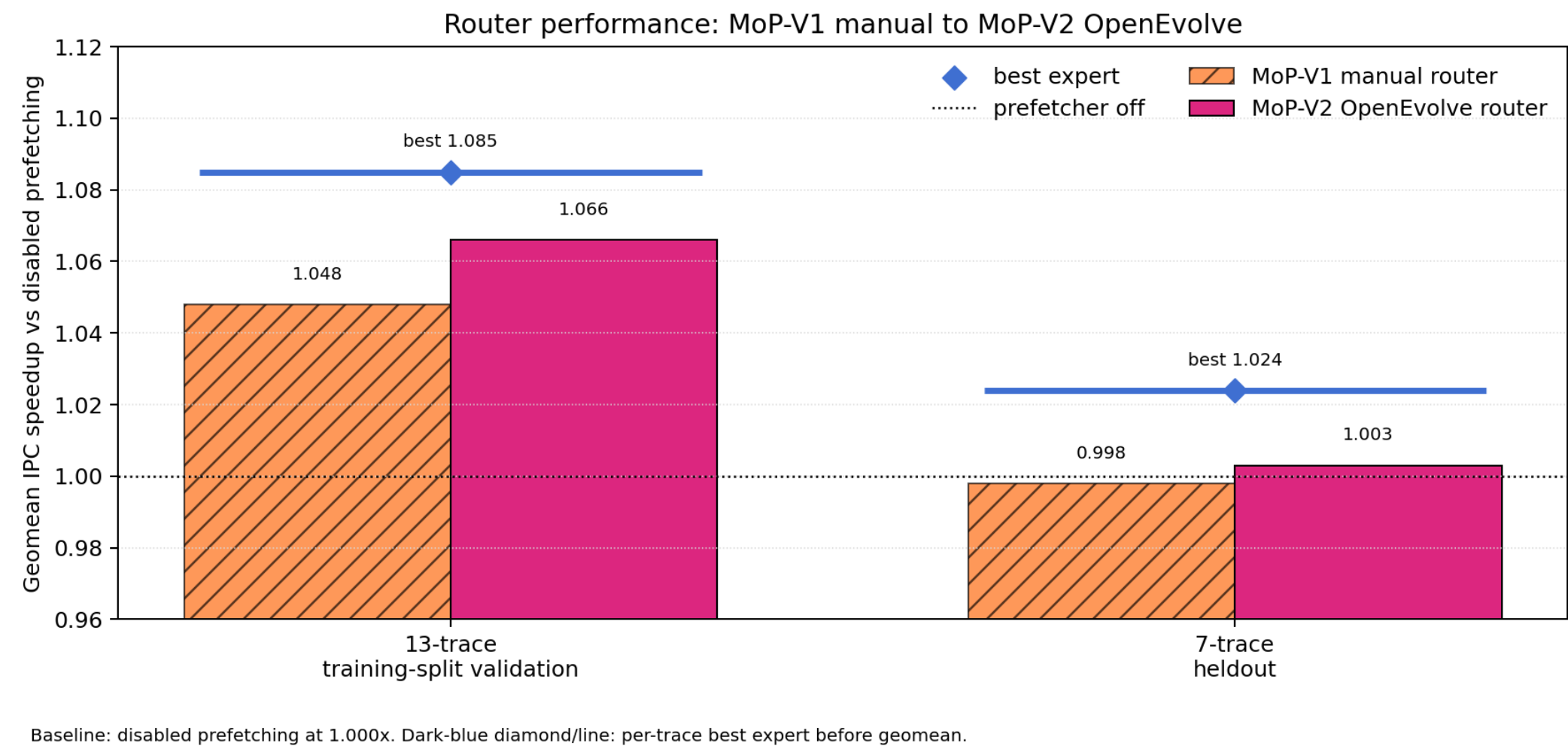
24 traces: SPEC, PARSEC, Ligra, secret_compute.
- Training

17 for search. Final validation used 13 available traces.
- Heldout

7 frozen traces after policy selection.
- Window

Validation 500K + 1M. Heldout 20M + 50M.

Before and after OpenEvolve

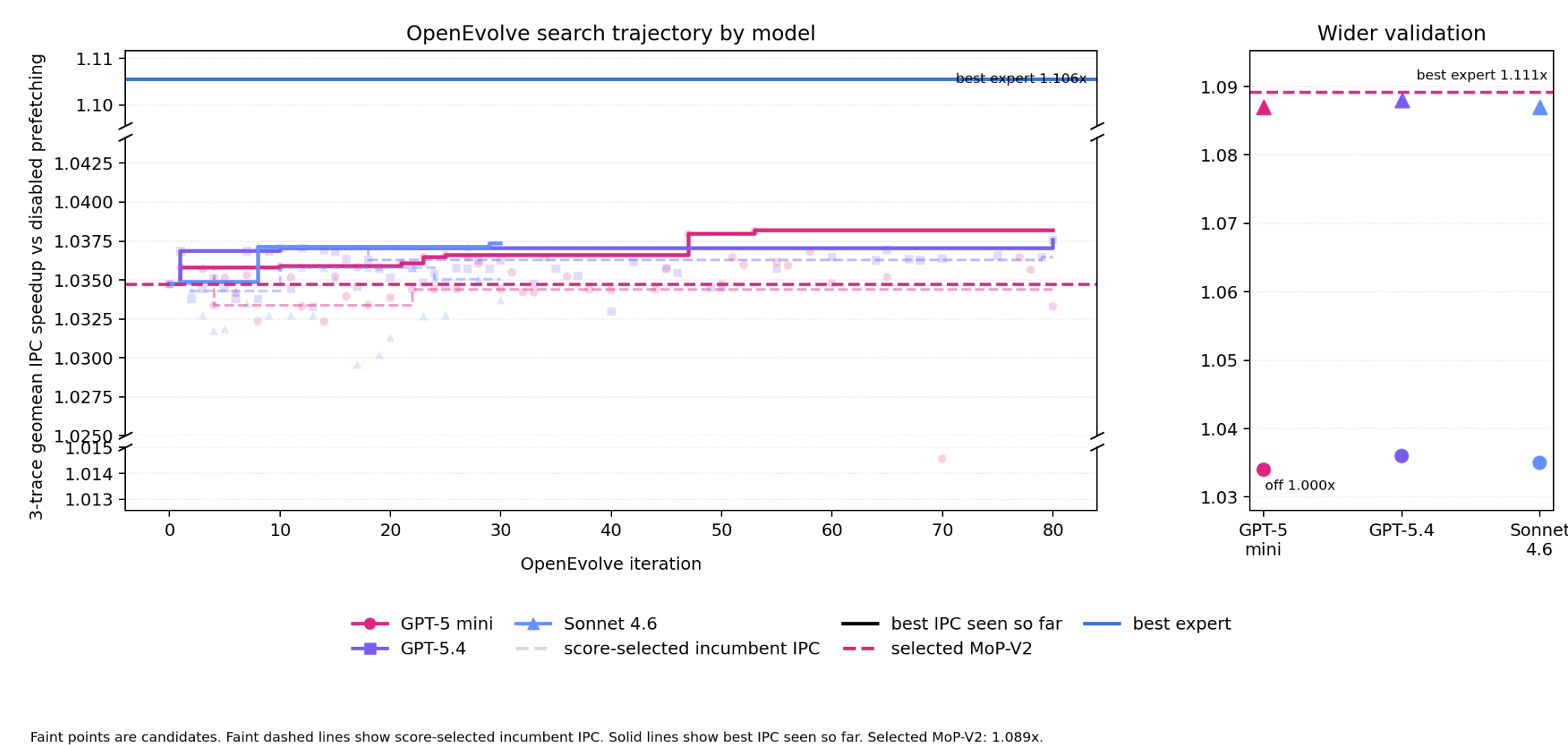


Caption: bars compare disabled prefetching, each single expert, the manual MoP-V1 router, the OpenEvolve-tuned MoP-V2 router, and the best expert reference on the same heldout and training-validation surfaces.

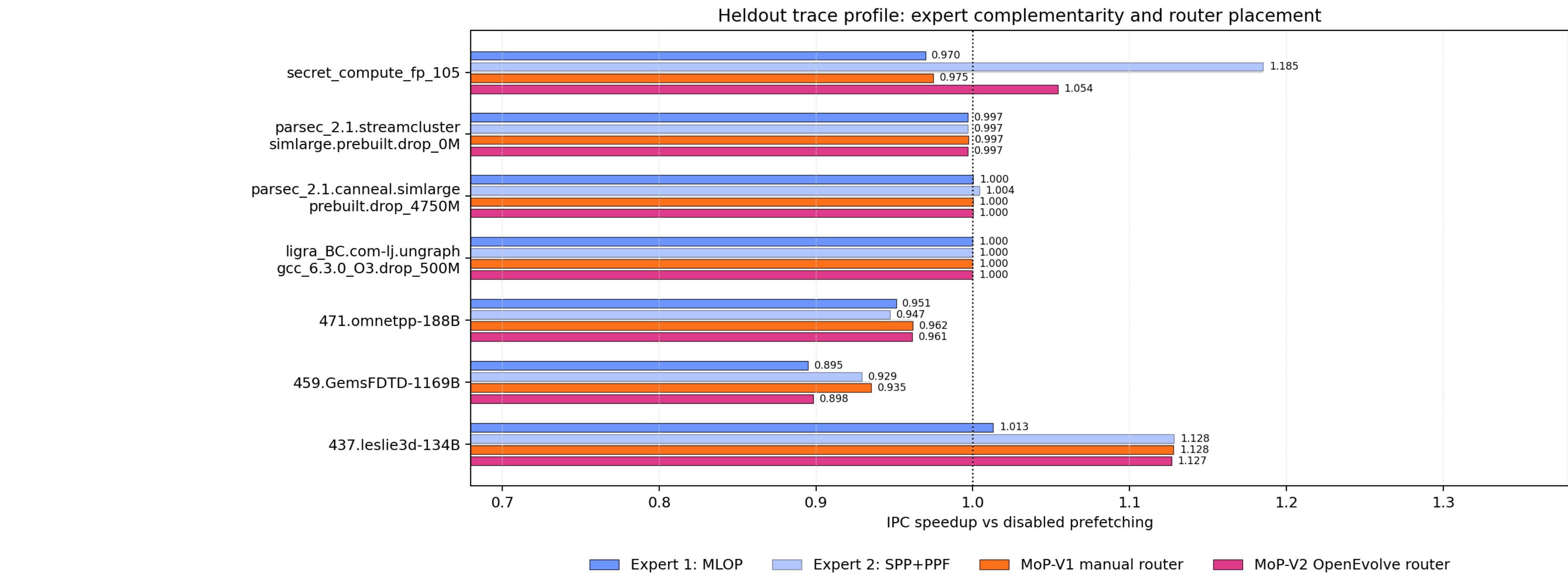
Heldout trace profile

Caption: each row is a heldout trace. Bars use disabled prefetching as 1.000x. MLOP and SPP+PPF sit above the MoP bars so the reader can see when the two experts disagree and where the router lands.

Search trajectory by model



Caption: faint points are generated candidates. Solid traces show best IPC seen so far. Dashed lines show the score-selected incumbent IPC.



Each trace row is ordered top-to-bottom: Expert 1, Expert 2, MoP-V1, and MoP-V2. Black dotted line is disabled prefetching at 1.000x.